Lecture_Scribe - Lecture Scribe Exam Prep

Tutorial_2_Solution.pdf
PDF

Generate a lecture scribe suitable for exam preparation based on the provided pdf for the context. While generating the Scribe, reason step by step through definitions, assumptions, proofs and example solutions exactly as presented in the provided context. Make sure you give the step by step derivation for the problems and example solutions. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce any material or derivation which is not present in the context.

Here is your **Lecture Scribe for Tutorial-2 (CSE 400: Fundamentals of Probability in Computing, SEAS, Ahmedabad University)** based strictly on the provided solution document .

All derivations, logic, and steps follow exactly the structure given in the PDF. No additional material has been introduced.

📘 Lecture Scribe – Tutorial 2 Solutions

Course: **CSE 400 – Fundamentals of Probability in Computing**
Institution: **School of Engineering and Applied Science (SEAS), Ahmedabad University**

Q1. Random Point on a Line Segment

Problem Interpretation

A point is chosen uniformly at random on a segment of length L.

Let:

- X = distance from one end

- $0 < x < L$

The two segments formed are:

- x
- $L - x$

The ratio of shorter to longer segment:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We need:

$$P(R < 1/4)$$

Step 1: Divide Based on Midpoint

Since shorter segment depends on position relative to midpoint:

Case 1: $x \leqslant L/2$

Shorter = x
Longer = $L - x$

Condition:

$$\frac{x}{L - x} < \frac{1}{4}$$

Multiply both sides:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter = $L - x$
Longer = x

$$\frac{L - x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Step 2: Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Total length:

$$\frac{L}{5} + (L - \frac{4L}{5}) = \frac{2L}{5}$$

Since uniform over L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Classification with Normal Densities

Given:

- $X \mid W \sim N(4, 4) \Rightarrow \sigma_1 = 2$
- $X \mid B \sim N(6, 9) \Rightarrow \sigma_2 = 3$
- $P(B) = \alpha, P(W) = 1 - \alpha$
- Observed: $X = 5$

We want equal error probabilities:

$$P(B \mid X = 5) = P(W \mid X = 5) = 1/2$$

Step 1: Apply Bayes' Theorem

$$P(B \mid X = 5) = \frac{\alpha f(5 \mid B)}{\alpha f(5 \mid B) + (1 - \alpha) f(5 \mid W)} = 1/2$$

Step 2: Compute Densities

Normal density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/2\sigma^2}$$

$$f(5 \mid B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5 \mid W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = 1/2$$

Step 3: Solve

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Fire Station Location

(a) Uniform Distribution on (0, A)

Let:

$$X \sim \text{Uniform}(0, A)$$

We minimize:

$$E[\mid X - a \mid]$$

Break absolute value:

$$E[\mid X - a \mid] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

After evaluation and differentiation:

$$\frac{d}{da}\left(\frac{2a^2 - 2aA + A^2}{2A}\right) = \frac{1}{2A}(4a - 2A)$$

Set to zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

Station at midpoint.

(b) Exponential Distribution

$$X \sim \text{Exponential}(\lambda)$$

$$E[\,|X - a|\,] = \int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Derivative:

$$\frac{d}{da} E[\,|X - a|\,] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = 1/2$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Battery lifetime:

$$X > t$$

Total probability:

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Similarly:

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$P(X > t + s \mid X > t) = \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Poisson → Binomial

Given:

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Find:

$$P(X = k \mid X + Y = n)$$

$$= \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Substitute Poisson pmf:

After simplification:

$$= \binom{n}{k} \left(\frac{\lambda_1}{\lambda_1 + \lambda_2}\right)^k \left(\frac{\lambda_2}{\lambda_1 + \lambda_2}\right)^{n-k}$$

Hence Binomial distribution.

Q6. Transformation of Uniform Variable

Given:

$$X \sim Uniform[-2, 2]$$

Piecewise transformation defined.

Key Observations

- Continuous parts on (-2,-1) and (1,2)

- Point mass at 0

$$P(Y = 0) = 1/2$$

PDF:

$$f_Y(y) = \begin{cases} 1/4 & -2 < y < -1 \\ 1/4 & 1 < y < 2 \\ 0 & otherwise \end{cases}$$

with point mass 1/2 at 0.

Q7. Stock Model (Binomial Approximation)

Price:

$$S_{1000} = S_0 u^k d^{1000-k}$$

Require:

$$u^k d^{1000-k} \geqslant 1.30$$

Taking logs:

$$k(\ln u - \ln d) + 1000 \ln d \geqslant \ln(1.30)$$

Solving:

$$k \geqslant 469.4$$

$$k \geqslant 470$$

$$k \sim Binomial(1000, 0.52)$$

Required probability:

$$P(k \geqslant 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$\lambda = 1/2$$

PDF:

$$f(x) = \frac{1}{2}e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geqslant 10 \mid X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

CDF:

$$F_Z(z) = P(X^2 + Y^2 \leqslant z^2)$$

Double integral over circle:

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate using Leibniz rule:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)\,[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})]}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \ldots, X_n)$$

$$P(M_n \leqslant x) = F(x)^n$$

Two cases:

Case 1: $x \leqslant y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(y)^{n+1}$$

✅ End of Tutorial-2 Lecture Scribe

If you'd like, I can now:

- Convert this into a **condensed 2-page exam revision sheet**
- Create a **formula + concept checklist**
- Or prepare **practice exam-style questions** from this tutorial.

Generate a lecture scribe suitable for exam preparation based on the provided pdf for the context. While generating the Scribe, reason step by step through definitions, assumptions, proofs and example solutions exactly as presented in the provided context. Make sure you give the step by step derivation for the problems and example solutions. Make the logical flow explicit so that a student can follow the reasoning during revision. Do not introduce any material or derivation which is not present in the context.

Here is your **exam-ready Lecture Scribe** based strictly and completely on the provided document .
All definitions, assumptions, case splits, derivations, and logical transitions are reproduced exactly as presented. No additional theory or shortcuts are introduced.

📘 CSE 400 – Fundamentals of Probability in Computing

Tutorial-2 Lecture Scribe (Step-by-Step Derivations)

Institution: School of Engineering and Applied Science (SEAS), Ahmedabad University
Source: Tutorial-2 Solution

Q1. Random Point on a Line Segment

Step 1: Interpretation of "Randomly Chosen"

A point is chosen uniformly on a segment of length L.

Let:

$$0 < x < L$$

be the distance from one endpoint.

Two segments formed:

- Length x
- Length $L - x$

Ratio:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We need:

$$P(R < 1/4)$$

Step 2: Logical Case Split at Midpoint

Because the shorter segment depends on whether x is left or right of $L/2$:

Case 1: $x \leqslant L/2$

Shorter = x
Longer = $L - x$

Condition:

$$\frac{x}{L - x} < \frac{1}{4}$$

Multiply:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter = $L - x$

Longer = x

$$\frac{L - x}{x} < \frac{1}{4}$$

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Step 3: Valid Region

$$(0, L/5) \cup (4L/5, L)$$

Length:

$$(\frac{L}{5} - 0) + (L - \frac{4L}{5}) = \frac{2L}{5}$$

Uniform selection over total length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Decision with Normal Densities

Given

White region:

$$X \mid W \sim N(4, 4)$$

$$\sigma_1 = 2$$

Black region:

$$X \mid B \sim N(6, 9)$$

$$\sigma_2 = 3$$

Priors:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observation:

$$X = 5$$

Requirement: Equal probability of error

$$P(B \mid X = 5) = P(W \mid X = 5) = \frac{1}{2}$$

Step 1: Apply Bayes' Theorem

$$P(B \mid X = 5) = \frac{\alpha f(5 \mid B)}{\alpha f(5 \mid B) + (1 - \alpha) f(5 \mid W)} = \frac{1}{2}$$

Step 2: Substitute Normal Density

Normal density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Compute:

$$f(5 \mid B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

$$f(5 \mid W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Step 3: Solve for α

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Fire Station Location

(a) Uniform Distribution on (0,A)

Let:

$$X \sim Uniform(0, A)$$

PDF:

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[\,|X - a|\,]$$

Break absolute value:

$$E[\,|X - a|\,] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

After evaluation:

$$E[\,|X - a|\,] = \frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{d}{da} = \frac{1}{2A}(4a - 2A)$$

Set zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

Station at midpoint.

(b) Exponential Distribution

$$X \sim Exponential(\lambda)$$

PDF:

$$\lambda e^{-\lambda x}$$

$$E[\mid X - a \mid] = \int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Derivative:

$$\frac{d}{da} E[\mid X - a \mid] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponentials

Let X = lifetime.

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Conditional probability:

$$P(X > t + s \mid X > t) = \frac{p_1 e^{-\lambda_1(t+s)} + p_2 e^{-\lambda_2(t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Poisson Conditional Distribution

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Find:

$$P(X = k \mid X + Y = n)$$

$$= \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Substitute pmf:

$$= \frac{(\frac{e^{-\lambda_1}\lambda_1^k}{k!})(\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!})}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$$

Simplify:

$$= \binom{n}{k}(\frac{\lambda_1}{\lambda_1 + \lambda_2})^k (\frac{\lambda_2}{\lambda_1 + \lambda_2})^{n-k}$$

Q6. Transformation of Uniform Variable

$$X \sim Uniform[-2, 2]$$

Piecewise transformation:

- $x \in [-2, -1] \Rightarrow Y = x$
- $(-1, 1) \Rightarrow Y = 0$
- $[1, 2] \Rightarrow Y = x$

$$f_X(x) = \frac{1}{4}$$

CDF Construction

For $y < -2$

$$F_Y(y) = 0$$

For $-2 \leqslant y < -1$

$$F_Y(y) = \frac{y+2}{4}$$

For $-1 \leqslant y < 0$

$$F_Y(y) = \frac{1}{4}$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

$$F_Y(0) = \frac{3}{4}$$

For $0 < y < 1$

$$F_Y(y) = \frac{3}{4}$$

For $1 \leqslant y \leqslant 2$

$$F_Y(y) = \frac{y+2}{4}$$

For $y > 2$

$$F_Y(y) = 1$$

PDF

For $(-2, -1)$ and $(1, 2)$:

$$f_Y(y) = \frac{1}{4}$$

Point mass at 0:

$$P(Y = 0) = \frac{1}{2}$$

Q7. Stock Price Model

After 1000 periods:

$$S_{1000} = S_0 u^k d^{1000-k}$$

Condition:

$$u^k d^{1000-k} \geqslant 1.30$$

Take log:

$$k \ln u + (1000 - k) \ln d \geqslant \ln(1.30)$$

$$k(\ln u - \ln d) + 1000 \ln d \geqslant \ln(1.30)$$

$$k \geqslant \frac{\ln(1.30) - 1000 \ln d}{\ln u - \ln d}$$

Substitute:

$$k \geqslant 469.4$$

$$k \geqslant 470$$

$$k \sim Binomial(1000, 0.52)$$

$$P(k \geqslant 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$\lambda = \frac{1}{2}$$

PDF:

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geqslant 10 \mid X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leqslant z^2)$$

Joint density:

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Region:

$$-z \leqslant x \leqslant z$$

$$-\sqrt{z^2 - x^2} \leqslant y \leqslant \sqrt{z^2 - x^2}$$

$$F_Z(z) = \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate using Leibniz rule:

$$\frac{d}{dz} \sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final PDF:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x) \left[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})\right]}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \ldots, X_n)$$

$$P(M_n \leqslant x) = F(x)^n$$

Case 1: $x \leqslant y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(y)^{n+1}$$

✅ End of Tutorial-2 Complete Exam Scribe

If you want, I can now generate:

- A **compressed 5-page rapid revision version**
- A **concept-mapping sheet**
- Or **exam-style derivation questions with marking scheme**

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Here is your **complete, structured, step-by-step Lecture Scribe for exam preparation**, based strictly on the provided document .

All reasoning, assumptions, case splits, integrals, algebraic steps, and conclusions are reproduced exactly as presented in the PDF. No additional concepts or shortcuts are introduced.

📘 CSE 400 – Fundamentals of Probability in Computing

Tutorial–2: Detailed Lecture Scribe for Revision

(SEAS, Ahmedabad University)

Q1. Random Point on a Line Segment

Interpretation

A point is chosen **uniformly at random** on a line segment of length L.

Let:

$$0 < x < L$$

be the distance from one endpoint.

Two segments formed:

- x
- $L - x$

Define the ratio:

$$R = \frac{\min(x, L - x)}{\max(x, L - x)}$$

We want:

$$P(R < 1/4)$$

Step 1: Logical Case Division at the Midpoint

The shorter segment depends on whether the point lies before or after $L/2$.

Case 1: $x \leqslant L/2$

Shorter = x
Longer = $L - x$

Condition:

$$\frac{x}{L-x} < \frac{1}{4}$$

Multiply both sides:

$$4x < L - x$$

$$5x < L$$

$$x < \frac{L}{5}$$

Case 2: $x > L/2$

Shorter = $L - x$
Longer = x

$$\frac{L-x}{x} < \frac{1}{4}$$

Multiply:

$$4L - 4x < x$$

$$5x > 4L$$

$$x > \frac{4L}{5}$$

Step 2: Determine Valid Region

Valid x:

$$(0, L/5) \cup (4L/5, L)$$

Length of region:

$$(\frac{L}{5} - 0) + (L - \frac{4L}{5}) = \frac{L}{5} + \frac{L}{5} = \frac{2L}{5}$$

Since selection is uniform over length L:

$$P = \frac{2L/5}{L} = \frac{2}{5}$$

Q2. Bayesian Decision Rule with Normal Distributions

Given

White region:

$$X \mid W \sim N(4, 4)$$

$$\sigma_1 = 2$$

Black region:

$$X \mid B \sim N(6, 9)$$

$$\sigma_2 = 3$$

Priors:

$$P(B) = \alpha, \quad P(W) = 1 - \alpha$$

Observation:

$$X = 5$$

Requirement:
Equal error probability:

$$P(B \mid X = 5) = P(W \mid X = 5) = \frac{1}{2}$$

Step 1: Apply Bayes' Theorem

$$P(B \mid X = 5) = \frac{\alpha f(5 \mid B)}{\alpha f(5 \mid B) + (1 - \alpha) f(5 \mid W)} = \frac{1}{2}$$

Step 2: Compute Likelihoods

Normal density:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

For Black:

$$f(5 \mid B) = \frac{1}{3\sqrt{2\pi}} e^{-1/18}$$

For White:

$$f(5 \mid W) = \frac{1}{2\sqrt{2\pi}} e^{-1/8}$$

Substitute:

$$\frac{3\alpha}{3\alpha + 2(1-\alpha)e^{-5/72}} = \frac{1}{2}$$

Step 3: Solve for α

$$6\alpha = 3\alpha + 2(1-\alpha)(0.93)$$

$$3\alpha = 1.86 - 1.86\alpha$$

$$4.86\alpha = 1.86$$

$$\alpha \approx 0.3827$$

Q3. Fire Station Location

(a) Uniform Distribution on (0,A)

$$X \sim Uniform(0, A)$$

$$f_X(x) = \frac{1}{A}$$

Minimize:

$$E[\mid X - a \mid]$$

Break integral:

$$E[\mid X - a \mid] = \int_0^a (a - x)\frac{1}{A}dx + \int_a^A (x - a)\frac{1}{A}dx$$

Evaluating leads to:

$$E[\mid X - a \mid] = \frac{2a^2 - 2aA + A^2}{2A}$$

Differentiate:

$$\frac{d}{da} = \frac{1}{2A}(4a - 2A)$$

Set zero:

$$4a = 2A$$

$$a = \frac{A}{2}$$

Station at midpoint.

(b) Exponential Distribution

$$X \sim Exponential(\lambda)$$

$$f(x) = \lambda e^{-\lambda x}$$

$$E[\mid X - a \mid] = \int_0^a (a - x)\lambda e^{-\lambda x} dx + \int_a^\infty (x - a)\lambda e^{-\lambda x} dx$$

Derivative:

$$\frac{d}{da} E[\mid X - a \mid] = 1 - 2e^{-\lambda a}$$

Set zero:

$$2e^{-\lambda a} = 1$$

$$e^{-\lambda a} = \frac{1}{2}$$

$$a = \frac{\ln 2}{\lambda}$$

Q4. Mixture of Exponential Lifetimes

Two battery types:

Type i:

$$X \mid T = i \sim Exponential(\lambda_i)$$

$$P(T = i) = p_i$$

Step 1: Total Probability

$$P(X > t) = p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}$$

Similarly:

$$P(X > t + s) = p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}$$

Step 2: Conditional Probability

$$P(X > t + s \mid X > t) = \frac{P(X > t + s)}{P(X > t)}$$

$$= \frac{p_1 e^{-\lambda_1 (t+s)} + p_2 e^{-\lambda_2 (t+s)}}{p_1 e^{-\lambda_1 t} + p_2 e^{-\lambda_2 t}}$$

Q5. Conditional Distribution of Poisson Variables

$$X \sim Poisson(\lambda_1)$$

$$Y \sim Poisson(\lambda_2)$$

Find:

$$P(X = k \mid X + Y = n)$$

Step 1: Definition

$$= \frac{P(X = k, Y = n - k)}{P(X + Y = n)}$$

Step 2: Substitute pmf

$$= \frac{\left(\frac{e^{-\lambda_1}\lambda_1^k}{k!}\right)\left(\frac{e^{-\lambda_2}\lambda_2^{n-k}}{(n-k)!}\right)}{\frac{e^{-(\lambda_1+\lambda_2)}(\lambda_1+\lambda_2)^n}{n!}}$$

Step 3: Simplify

Collect terms:

$$= \binom{n}{k}\left(\frac{\lambda_1}{\lambda_1+\lambda_2}\right)^k\left(\frac{\lambda_2}{\lambda_1+\lambda_2}\right)^{n-k}$$

Q6. Transformation of Uniform Variable

$$X \sim Uniform[-2, 2]$$

$$f_X(x) = \frac{1}{4}$$

Transformation:

- $x \in [-2, -1] \Rightarrow Y = x$
- $(-1, 1) \Rightarrow Y = 0$
- $[1, 2] \Rightarrow Y = x$

CDF Construction

- $y < -2$: 0
- $-2 \leqslant y < -1$: $\frac{y+2}{4}$
- $-1 \leqslant y < 0$: $\frac{1}{4}$
- At 0: jump of $\frac{1}{2}$
- $0 < y < 1$: $\frac{3}{4}$
- $1 \leqslant y \leqslant 2$: $\frac{y+2}{4}$
- $y > 2$: 1

PDF

$$f_Y(y) = \frac{1}{4} \quad \text{for } (-2, -1) \text{ and } (1, 2)$$

Point mass:

$$P(Y = 0) = \frac{1}{2}$$

Q7. Stock Price Model

$$S_{1000} = S_0 u^k d^{1000-k}$$

Require:

$$u^k d^{1000-k} \geqslant 1.30$$

Take logs:

$$k(\ln u - \ln d) + 1000 \ln d \geqslant \ln(1.30)$$

Solve:

$$k \geqslant 469.4$$

$$k \geqslant 470$$

$$k \sim Binomial(1000, 0.52)$$

$$P(k \geqslant 470) = \sum_{k=470}^{1000} \binom{1000}{k} (0.52)^k (0.48)^{1000-k}$$

Q8. Exponential Repair Time

$$\lambda = \frac{1}{2}$$

$$f(x) = \frac{1}{2} e^{-x/2}$$

(a)

$$P(X > 2) = e^{-1}$$

(b)

$$P(X \geqslant 10 \mid X > 9) = \frac{e^{-5}}{e^{-9/2}} = e^{-1/2}$$

Q9. Distribution of $Z = \sqrt{X^2 + Y^2}$

$$F_Z(z) = P(X^2 + Y^2 \leqslant z^2)$$

$$= \int_{-z}^{z} \int_{-\sqrt{z^2-x^2}}^{\sqrt{z^2-x^2}} f_X(x) f_Y(y) dy dx$$

Differentiate:

$$\frac{d}{dz}\sqrt{z^2 - x^2} = \frac{z}{\sqrt{z^2 - x^2}}$$

Final:

$$f_Z(z) = z \int_{-z}^{z} \frac{f_X(x)\,[f_Y(\sqrt{z^2 - x^2}) + f_Y(-\sqrt{z^2 - x^2})]}{\sqrt{z^2 - x^2}} dx$$

Q10. Joint Distribution of M_n and M_{n+1}

$$M_n = \max(X_1, \dots, X_n)$$

$$P(M_n \leqslant x) = F(x)^n$$

Case 1: $x \leqslant y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(x)^n F(y)$$

Case 2: $x > y$

$$P(M_n \leqslant x, M_{n+1} \leqslant y) = F(y)^{n+1}$$

✅ End of Complete Step-by-Step Lecture Scribe

If you would like, I can now create:

- A **condensed derivation-only quick revision sheet**
- A **formula flow chart**
- Or a **mock exam with full worked solutions following this exact style**